Google Workspace Engineer Handbook

Administrator → Engineer → Security Engineer → Architect

Version: 1.0

About this Handbook

This handbook is designed to take a learner from a Google Workspace Administrator to a Google Workspace Engineer, Security Engineer, and finally an Architect. It combines conceptual knowledge, enterprise design principles, troubleshooting methodologies, security best practices, migration strategies, and interview preparation.

Every chapter follows the same structure:

1. Business Problem
2. Why the Feature Exists
3. Architecture
4. How It Works
5. Enterprise Example
6. Comparison Tables
7. Best Practices
8. Common Mistakes
9. Troubleshooting
10. Interview Questions
11. Summary

Learning Status

Completed Topics

Module 1 – Google Workspace Fundamentals

Topics Covered

- Google Workspace Overview
- Editions
- Tenant Architecture
- Core Services
- Licensing

- Admin Console Overview

Concepts Learned

- Google Workspace is an identity-centric SaaS platform.
- A tenant represents an organization's Google Workspace environment.
- Licensing determines available features.
- Identity is the foundation of all Workspace services.

Enterprise Example

Company A has 15,000 employees.

One Workspace tenant contains:

- Users
- OUs
- Groups
- Shared Drives
- Gmail
- Meet
- Drive
- Calendar
- Security Policies

Everything is centrally managed.

Module 2 – Identity Management

Topics Covered

- Users
- Organizational Units
- Groups
- Roles
- Admin Roles
- Privileges
- Secondary Domains
- Domain Alias
- Delegated Administration

Example

Engineering OU

- Developers
- QA

- DevOps

HR OU

- HR Managers
- Recruiters

Each OU receives different security policies.

Comparison

OU	Group
Used for policy management	Used for collaboration
User belongs to only one OU	User can belong to many Groups
Controls settings	Controls access and communication

Module 3 – Authentication & Identity

Topics Covered

- Authentication
- Authorization
- Identity
- SAML
- OAuth
- OpenID Connect
- JWT
- Access Token
- ID Token
- Refresh Token
- Digital Signatures

Comparison

Authentication	Authorization
Verifies identity	Determines permissions
Password	Roles
SAML	OAuth
MFA	IAM

Remember

Authentication answers:

Who are you?

Authorization answers:

What are you allowed to do?

Module 4 – SAML

Topics Covered

- SAML Flow
- Assertion
- NameID
- Entity ID
- ACS URL
- Certificates
- Signature Validation
- SP Initiated
- IdP Initiated

Enterprise Scenario

Google Workspace

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Redirect

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Okta

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User Authentication

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SAML Assertion

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Google Validates Signature

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User Logged In

Troubleshooting

- Invalid Certificate
- Incorrect ACS URL
- NameID mismatch
- Clock skew
- Assertion expired

Module 5 – OAuth & OIDC

Topics Covered

- OAuth Flow
- Consent Screen
- Scopes
- Tokens
- OIDC
- JWT

Comparison

OAuth	OIDC
Authorization	Authentication
Access Token	ID Token
API Access	User Identity

Enterprise Example

CRM Application

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Requests Drive Access

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User Grants Consent

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Google Issues Access Token

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CRM Accesses Drive APIs

Module 6 – SCIM Provisioning

Topics Covered

- SCIM
- Provisioning
- Attribute Mapping
- JIT
- Duplicate Accounts
- Provisioning Errors

Comparison

SCIM	JIT
Creates users automatically	Creates users during first login
Lifecycle management	Login based
Supports updates	Limited

Common Issues

- Duplicate users
- Attribute mismatch
- Authorization failure
- Existing Google account conflict

Module 7 – Gmail Administration

Topics Covered

- MX
- SPF
- DKIM
- DMARC
- SMTP
- Mail Flow
- Routing
- Email Security

Comparison

Record	Purpose
MX	Mail Delivery
SPF	Sender Validation
DKIM	Message Integrity
DMARC	Policy Enforcement

Mail Flow

Internet

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MX

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Google Mail Server

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Spam Check

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Mailbox

Module 8 – Google Drive

Topics Covered

- Shared Drives
- Ownership
- Permissions
- Trusted Domains
- External Sharing
- Audit Logs

Comparison

My Drive	Shared Drive
User owns files	Organization owns files
Ownership changes	Ownership remains with organization
User departure affects files	Business continuity maintained

Module 9 – Migration

Topics Covered

- Discovery
- Assessment
- Pilot
- Cutover
- Phased Migration
- Coexistence
- Communication
- Validation

Migration Tools

- Data Migration Service
- Google Workspace Migrate
- GWMME
- GWMMO
- CloudM

Comparison

Tool	Use Case
DMS	Basic IMAP & Exchange migrations
GWMME	Exchange to Gmail
GWMMO	Outlook data migration
Google Workspace Migrate	Large enterprise migrations
CloudM	End-to-end migration and coexistence

Migration Process

Assess

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Pilot

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Validate

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Production

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Support

Module 10 – APIs & Automation

Topics Covered

- Service Accounts
- Domain-Wide Delegation
- Admin SDK
- Directory API
- Least Privilege

Design Principle

One service account

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One business function

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Minimum required scopes

Module 11 – Enterprise Troubleshooting

Framework

1. Authentication
2. Authorization
3. Context
4. Provisioning

5. Application

Always ask:

Which layer is failing?

Module 12 – Security Center

Topics Covered

- Dashboard
- Security Health
- Recommendations
- Security Posture

Purpose

Understand overall organizational security.

Module 13 – Investigation Tool

Topics Covered

- Admin Logs
- Gmail Logs
- Drive Logs
- Calendar Logs
- OAuth Logs

Workflow

Incident

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Search

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Investigate

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Root Cause

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Remediation

Module 14 – Context-Aware Access

Topics Covered

- Zero Trust
- Access Levels
- Managed Devices
- Endpoint Verification
- Location Based Access

Comparison

SAML	Context-Aware Access
Authenticates user	Evaluates access context
Identity	Device + Location + Risk

Module 15 – Data Loss Prevention

Topics Covered

- Gmail DLP
- Drive DLP
- Detectors
- Conditions
- Actions
- Exceptions

Design Principle

Monitor

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Warn

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Block

Module 16 – OAuth Security

Topics Covered

- OAuth Risks
- Scopes
- App Access Control
- API Controls
- Token Abuse

Security Principle

Review permissions,

not just applications.

Module 17 – Domain-Wide Delegation Security

Topics Covered

- DWD
- Service Accounts
- Scope Reviews
- Least Privilege
- Separation of Duties

Design Principle

Never create a "super service account."

Separate service accounts by business function.

Remaining Topics

Google Vault

- Holds
- Matters
- Retention
- eDiscovery
- Export
- Compliance

Endpoint Management

- BYOD
- Company Devices
- Device Compliance
- Remote Wipe
- Chrome Enterprise
- Windows & macOS Management

Enterprise Architecture

- Large Tenant Design
- OU Strategy
- Group Strategy
- Shared Drive Design
- Multi-Domain Architecture
- Delegated Administration

Advanced Automation

- Python
- Reports API
- Gmail API
- Drive API
- Cloud Identity API
- Automation Projects

Incident Response

- OAuth Compromise
- Insider Threat
- Data Exfiltration
- Mail Routing Failure
- SAML Failure
- SCIM Failure
- Executive Escalations

Compliance

- GDPR
- HIPAA

- ISO 27001
- SOC 2
- Audit Preparation

Mock Interviews

- Administrator
- Engineer
- Security Engineer
- Architect
- Consultant

Guiding Principles Learned

- Authentication answers "Who are you?"
- Authorization answers "What can you access?"
- Zero Trust means "Never trust, always verify."
- Apply the Principle of Least Privilege.
- Design for business continuity.
- Separate duties to reduce blast radius.
- Understand the business requirement before implementing security controls.
- Use structured troubleshooting instead of guessing.
- Think like an engineer: design for failure, compromise, and scale.

Personal Goal

Progress from:

Google Workspace Administrator

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Google Workspace Engineer

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Google Workspace Security Engineer

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Google Workspace Architect